

Benjamin Berman

SOFTWARE ENGINEER & TECHNICAL LEAD

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PROFESSIONAL SUMMARY

I am a software engineer and technical lead with a Computer Science Ph.D. and 11+ years of industry experience, including financial systems at BNY and Interactive Brokers. Java and Spring Boot are my bread and butter, but my broader focus is making complex systems reliable, understandable, and easier to change. I am an enthusiastic, pragmatic adopter of AI-assisted development, using coding agents within a development process grounded in static typing, automated testing, and human review to accelerate delivery without giving up correctness. I also enjoy mentoring developers and creating tools and workflows that help both human and AI contributors work effectively.

TECHNICAL SKILLS

Core: Java, Spring Boot, Spring MVC, Spring WebFlux, Spring Data JPA, microservices, REST, JSON, Java EE, Hibernate/JPA, Apache ActiveMQ Artemis, Kafka, Kafka Streams, Project Reactor

Data: SQL, Oracle/PL/SQL, PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Flyway, Jackson

Testing & Delivery: AI-assisted development, end-to-end testing, JUnit, TestNG, Git, GitLab CI/CD, Gradle, Maven, Docker, Kubernetes, Terraform, Ansible, Linux/Unix, Bash/Zsh

Cloud, Security & Observability: AWS (CCP certified; EC2, EKS, MSK, S3, RDS, SQS, API Gateway, CloudWatch), OAuth, JWT, SSL/TLS, Kibana, Grafana, nginx, Tomcat, JBoss

Additional: Scala, Ruby, JavaScript, TypeScript, Angular, AngularJS, SOAP, XML, YAML, Ant, Mercurial, Subversion, Coq, Alloy, Lustre, ESC/Java2, Swing

PROFESSIONAL EXPERIENCE

eENGINE — Pittsburgh, PA | *Senior Software Engineer* — *BNY engagement* | June 2025–Present

- Extended an asynchronous integration between BNY's Spring Boot/Angular digital-assets platform and its enterprise information warehouse to support financial reporting; implemented and tested local messaging workflows with Apache ActiveMQ Artemis.
 - Flagged the blast-radius risk of enabling reporting across approximately 60 transaction types, partnered with Product to define the initial scope, built AI-assisted end-to-end test coverage, and implemented per-transaction-type configuration to support incremental rollout.
 - Adapted quickly to a major late-stage requirement for authoritative system-of-record identifiers, working with CIW and Product stakeholders to refocus reporting on completed, fee-bearing blockchain transactions.
 - Updated the Oracle database schema and indexes to associate off-chain platform records with corresponding on-chain activity.
- Built an AI-assisted, multi-repository development environment with local service orchestration, end-to-end testing, parallel builds, and Bash/Zsh tab completion; optional local-database support reduced application startup from over two minutes to under 20 seconds. Presented the environment to other engineers and helped colleagues adopt it.

TeleTracking Technologies Inc — Pittsburgh, PA | *Software Engineer III* | July 2024–March 2025

- Debugged and fixed hospital logistics software on an event-oriented platform written in Java, C#, and Python.

UPMC Enterprises — Pittsburgh, PA | *Senior Software Engineer* | February 2022–April 2024

Contributed to a large-scale medical-records aggregation and analysis platform spanning a wide range of microservices and web applications.

- Used reactive streams to improve throughput by more than 3×.
- Debugged the CI/CD pipeline to improve some job completion times by up to 8×.

M*Modal / 3M — Pittsburgh, PA | *Senior Software Engineer; included service as tech lead* | July 2017–February 2022

Developed Java/Spring Boot services with architectural, leadership, and mentoring responsibilities for Computer Assisted Provider Documentation (CAPD), which connected a physician-facing interface to real-time natural-language processing.

- Designed and implemented Kafka Streams microservices that processed and aggregated approximately 1,000 requests per second in real time; coordinated across application, reporting, legacy ETL, product, and sales teams to resolve critical ambiguities.
- Connected real-time NLP with prior document analysis, handling identifier mappings, configuration and credential caching, OAuth, fallback mechanisms, additional API calls, and observability.
- Created reusable Terraform and Ansible automation for consistent deployment across AWS environments.

Interactive Brokers Group — Greenwich, CT | *Secure Application Developer* | February 2015–July 2017

- Developed secure financial applications using Java EE and Oracle/PL/SQL, with an emphasis on encryption and multi-factor authentication.

The University of Iowa — Iowa City, IA | *Research/Teaching Assistant and Instructor* | August 2008–December 2014

- Taught and developed material for introductory Java; supported undergraduate and graduate courses including HCI and Formal Methods.
- Developed and evaluated user interfaces for programming tools and helped secure NSF grant CCF-1250306.

EDUCATION

Ph.D., Computer Science — The University of Iowa, December 2014

B.S., Physics — Rice University, May 2008

PUBLICATIONS & PRESENTATIONS

Berman, B. A. and Hourcade, J. P. (2014). “Keyboard Card Menus: A New Presentation of Non-Standard Shortcuts.” *Journal of Universal Computer Science*.

Berman, B. A. and Hourcade, J. P. (2014). “Keyboard Card Menus: Faster Learning of Many Fast Commands.” XIV International Congress of Human-Computer Interaction (Interaction 2013).

“New Coq User Interfaces: Survey and Ideas.” Co-authored presentation, Coq Workshop 2012, Princeton, NJ.